



Robust respiratory and heart rate estimation via FMCW radar and sparse Bayesian adaptive filtering

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ABSTRACT

Non-contact vital sign detection, particularly for respiratory rate (RR) and heart rate (HR), shows significant potential in healthcare applications due to its comfort advantages and privacy preservation. Nevertheless, detecting weak heartbeat signals remains challenging because of respiratory harmonics, ambient noise, and environmental clutter. To address this challenge, we propose a vital signs detection system based on frequency-modulated continuous wave (FMCW) radar. The system is built around a streamlined framework that combines four core modules designed to accurately extract RR and HR from the radar signals. The proposed system begins by removing static clutter using a Moving Target Detection (MTD) algorithm within the preprocessing module. Following this, the vital sign extraction module performs noncoherent integration to identify the distance bins most relevant to physiological activity. In the signal enhancement module, Pearson correlation analysis is used to combine signal regions with high consistency, while first-order phase differencing is applied to suppress baseline drift. This improves the stability and reliability of the extracted signals. To isolate respiratory and heart signals, the system employs a wavelet packet decomposition with autocorrelation (WPD-AC) algorithm. This approach effectively separates the respiratory and heart components while reducing noise and suppressing harmonic interference. Finally, a sparse Bayesian least-mean-squares (SBLMS) algorithm is introduced to dynamically refine prior information. This enables more accurate estimation of RR and HR. The experimental results show that the system achieves mean absolute errors (MAE) of 0.53 Bpm for RR and 0.93 Bpm for HR, demonstrating its notable capability in suppressing noise and respiratory harmonic interference. Furthermore, extensive experiments conducted in realistic office environments show that, compared with existing methods, the system exhibits excellent robustness across different users, distances, and angles conditions.

1. Introduction

With an aging population and the increasing prevalence of chronic illnesses, non-contact vital signs detection has attracted significant attention for its potential in healthcare, emergency response, and smart home applications [1,2]. Traditional contact-based methods, such as electrocardiography (ECG) [3] and photoplethysmography (PPG) [4], offer high accuracy but require direct contact with the skin. This limits their use in cases involving burn victims or patients with infectious diseases. Non-contact techniques based on optical [5] or infrared [6] sensing address some of these issues but are often affected by ambient lighting, obstructions, and privacy concerns. In contrast, radar-based methods have emerged as a promising alternative due to their ability to penetrate clothing, maintain user privacy, provide high resolution, and operate reliably in various environments [7,8]. As a result, radar

technology is becoming a focal point in the development of non-contact vital signs detection systems.

Un-modulated continuous wave (CW) radar, impulse radio ultra-wideband (IR-UWB) radar, stepped frequency continuous wave (SFCW) radar, and frequency modulated continuous wave (FMCW) radar are the four main types of radar commonly used in vital signs detection [9]. CW radar has a simple structure and allows for continuous signal transmission and reception, making it suitable for real-time monitoring [10]. However, its fixed frequency prevents it from directly measuring the distance to a target, limiting its effectiveness in complex environments [11]. IR-UWB radar addresses this limitation by transmitting short Gaussian pulses in the time domain, which translates into a wide frequency bandwidth. This enables high distance resolution and obstacle penetration, making it well-suited for vital signs monitoring

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in cluttered or obstructed environments [12]. However, to enhance signal quality, IR-UWB systems must transmit multiple pulses, and the need for high-speed analog-to-digital converters (ADCs) adds to the system's cost and design complexity [13]. SFCW radar improves distance resolution by gradually varying the frequency of the transmitted signal [14], making it ideal for applications requiring precise range measurements. Nevertheless, it is affected by range-velocity coupling in moving targets, which complicates signal processing and results in slower scanning speeds [15]. FMCW radar overcomes these issues by using a linear frequency-modulated signal, which separates distance and velocity information. This decoupling allows for more accurate target detection. By mixing the received and transmitted signals, FMCW radar generates an intermediate frequency (IF) signal from which both distance and velocity can be extracted. This approach not only reduces system complexity and improves power efficiency but also maintains high range resolution and a strong signal-to-noise ratio. As a result, FMCW radar is particularly effective in detecting weak physiological signals, such as breathing and heartbeat signals, in non-contact detection scenarios [16,17].

Recent advancements in non-contact vital sign detection have introduced diverse methods to analyze and extract respiratory and heartbeat signals. One approach involves a dual-parameter least mean square (LMS) algorithm [18], which separates vital signs from mixed signals containing multiple respiratory harmonics, allowing for the simultaneous detection of breathing and heartbeat activity. Another method employs ensemble empirical mode decomposition (EEMD) [19], an enhancement of the traditional empirical mode decomposition (EMD) technique [20], to improve accuracy in extracting physiological signals. In addition, a non-recursive variational mode decomposition (VMD) algorithm was introduced in [21]. This technique iteratively identifies the optimal set of intrinsic mode functions (IMFs) to isolate respiratory and heartbeat components from time-series data. Wavelet transform-based methods [22] have also been explored, leveraging their ability to provide high resolution in both time and frequency domains for improved signal detection. Although existing methods have demonstrated a certain level of effectiveness in vital sign separation, they often exhibit limited robustness when dealing with non-stationary signals, primarily due to mode mixing and complex parameter dependencies. In practical environments, the thoracic micro-displacement induced by heartbeats is significantly smaller than that caused by respiration, making cardiac signals highly susceptible to being submerged in environmental clutter and background noise [23]. More critically, higher-order respiratory harmonics frequently overlap with heartbeat signals in the frequency domain, resulting in strong spectral coupling that makes it difficult for conventional filtering or decomposition techniques to effectively separate the two components. Given the inherently weak nature of heartbeat-induced signals, such overlap can easily cause the system to misinterpret dominant respiratory harmonics or noise peaks as the true heart rate, thereby severely degrading detection accuracy. Consequently, achieving robust and precise vital sign detection under practical conditions remains an open research question.

To solve the above problem, we propose a non-contact vital signs detection system based on FMCW radar in this paper. The system consists of four main modules: signal preprocessing, vital signs signal extraction, signal enhancement, and signal decomposition and reconstruction. First, the preprocessing module employs Moving Target Detection (MTD) to remove static clutter. Next, the vital signs extraction module performs noncoherent integration to identify range bins associated with physiological motion, enabling the isolation of vital signs signals. Following this, the signal enhancement module evaluates the similarity of neighboring displacement signals using Pearson correlation. Regions exhibiting frequency consistency are combined. To address residual phase baseline drift, a first-order phase differencing technique is employed. This step calculates the difference between consecutive phase values, improving signal continuity and highlighting phase variations associated with vital signs. Finally, the vital signs

signal decomposition and reconstruction module applies two algorithms: wavelet packet decomposition with autocorrelation (WPD-AC) algorithm and sparse Bayesian least mean square (SBLMS) algorithm. WPD-AC separates the respiration and heartbeat components into distinct frequency bands and reinforces their periodic patterns, effectively reducing noise and harmonic interference. Building on this, the SBLMS algorithm incorporates Bayesian sparse priors and an adaptive step-size mechanism to dynamically optimize signal reconstruction, further enhancing the detection accuracy of respiratory and heartbeat activity. The main contributions of this paper are as follows:

(1) The Pearson correlation-based phase consistency region fusion method is proposed. This method effectively enhances vital sign signals by performing a weighted fusion of highly correlated chest wall displacement signals from multiple range bins. On this basis, first-order phase difference and moving average filtering are further combined to suppress baseline drift and high-frequency noise, thereby significantly improving the signal-to-noise ratio. This strategy overcomes the limitation of traditional methods that rely on a single range bin for vital sign signal extraction.

(2) A cascaded framework integrating the WPD-AC algorithm and the SBLMS algorithm is designed for vital sign signal separation and enhancement. Specifically, the WPD-AC algorithm is first employed to separate respiratory and heartbeat components while suppressing environmental noise and respiratory harmonic interference. Subsequently, the SBLMS algorithm is introduced, which incorporates a Bayesian sparse prior and a variable step-size zero-attraction mechanism, resulting in sparser and more pronounced spectral peaks corresponding to heart rate and respiration rate. This approach effectively addresses the performance degradation of conventional adaptive filtering in sparse signal environments and mitigates the misdetection of heart rate caused by respiratory harmonics.

(3) We evaluated our method using data collected in a real-world office environment. The experimental results confirm that our approach effectively suppresses interference from respiratory harmonics, noise, and clutter, and significantly improves overall detection accuracy and robustness in complex environments compared to existing methods.

The remainder of this paper is structured as follows. Section 2 outlines the principles and methodology of FMCW radar-based vital sign detection. Section 3 describes the system design and implementation. Section 4 evaluates experimental results, demonstrating the system's efficacy. Finally, Section 5 concludes the paper.

2. Fmcw radar system model

The FMCW radar system operates by transmitting a series of linear frequency-modulated signals, known as chirps, through its antenna. The frequency of these chirps increases linearly over time, as shown in Fig. 1(a). The transmitted chirp signal can be expressed as:

$$x_T(t) = A_T \cos \left(2\pi f_s t + \pi \frac{B}{T_s} t^2 + \vartheta(t) \right) \quad (1)$$

where A_T denotes the transmitted signal amplitude, f_s is the starting frequency, B is the pulse bandwidth, T_s is the pulse duration, and $\vartheta(t)$ represents the phase noise of the transmitter.

When the transmitted signal encounters a human target or stationary object, it is reflected. The echo signal $x_R(t)$ is expressed as:

$$x_R(t) = A_R \cos \left(2\pi f_s(t - \tau) + \pi \frac{B}{T_s}(t - \tau)^2 + \vartheta(t - \tau) \right) \quad (2)$$

where A_R denotes the received signal amplitude, $\tau = 2R/c$ is the time delay between the transmitted and reflected signals, R is the radial distance between the target and the radar, and c is the speed of light.

The transmitted and received signals are coherently mixed via a mixer and undergo quadrature demodulation to produce the intermediate frequency (IF) signal, expressed as:

$$x_{IF}(t) = A_{IF} e^{j \left(2\pi \frac{B\tau}{T_s} t + 2\pi f_s \tau - \pi \frac{B\tau^2}{T_s} + \Delta\vartheta(t) \right)} \quad (3)$$

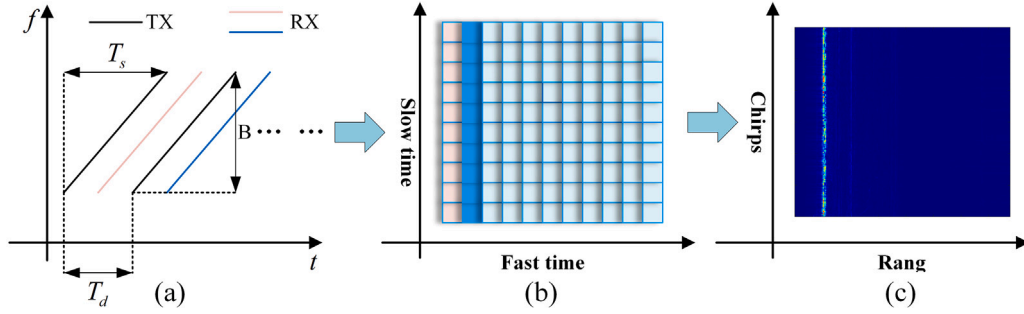


Fig. 1. FMCW radar system.

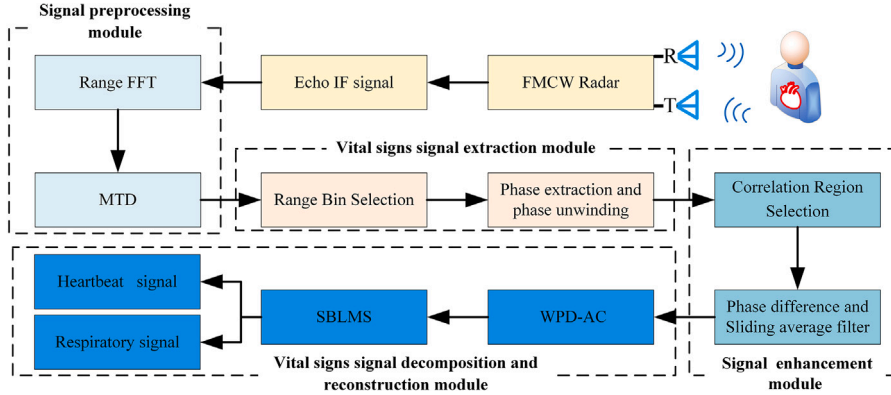


Fig. 2. Block diagram of the proposed signal processing chain.

where A_{IF} is the IF signal amplitude and $\Delta\theta(t) = \theta(t) - \theta(t - 2R/c)$ the residual signal noise, which can be ignored at short distances. In addition, due to the small displacement amplitude caused by respiration and heartbeat, $\pi \frac{B r^2}{T_s}$ can be ignored as well [1,24].

Ideally, the motion of the chest wall $x_0(t)$ can be regarded as the superposition of displacement component induced by the heartbeat $x_h(t)$ and displacement component caused by respiration $x_r(t)$. Consequently, the time-varying phase $\Delta\phi(t)$ is expressed as:

$$\begin{aligned} \Delta\phi(t) &= \frac{4\pi}{\lambda} R(t) = \frac{4\pi}{\lambda} x_0(t) + \frac{4\pi}{\lambda} x \\ &= \frac{4\pi}{\lambda} (x_h(t) + x_r(t)) + \frac{4\pi}{\lambda} x \end{aligned} \quad (4)$$

where $\lambda = c/f_s$ is the wavelength, x is the radial distance from a stationary human body to the radar. By analyzing the periodicity of the recovered chest wall displacement signals, we can accurately estimate respiration rate and heart rate.

However, in practical applications, signal recovery is subject to interference from various factors such as environmental and device noise. These noises can mask or distort the weak chest wall displacement signals, making it difficult to accurately capture their periodic characteristics. Moreover, chest wall displacement signals contain not only periodic displacements from respiration and heartbeats but may also include non-periodic movements such as minor body shifts or muscle tremors. These non-periodic components interfere with periodic signal analysis, increasing the difficulty of frequency estimation. This makes precise estimation of respiratory and heart rates through periodic signal analysis highly challenging.

3. Proposed method

In this paper, the block diagram of the signal processing chain for the proposed FMCW radar-based vital-sign detection method is shown in Fig. 2. It primarily consists of four functional modules: signal preprocessing, vital signs signal extraction, signal enhancement, and vital signs signal decomposition and reconstruction.

3.1. Signal preprocessing module

Effective vital sign extraction requires preprocessing of radar echo data. First, we apply Fast Fourier Transform (FFT) along the fast-time dimension to construct a 2D range-time matrix, leveraging the linear relationship between intermediate frequency signals and target distance, as shown in Fig. 1(b). This matrix captures chest vibrations caused by breathing and heartbeat, along with static environmental clutter. To filter out static clutter, MTD is used to identify chest movements, expressed as:

$$S'(t) = S(t) - \frac{1}{N} \sum_{n=1}^N S(t) \quad (5)$$

where $S(t)$ denotes the original echo matrix, $S'(t)$ is the echo matrix after static clutter removal, and N is the total number of slow-time samples. Fig. 3 demonstrates the clutter suppression efficacy. The original image is cluttered with noise, obscuring useful information. After processing, vital signs are clearly visible, facilitating subsequent signal extraction and analysis.

3.2. Vital signs signal extraction module

After eliminating clutter from the radar echo signals, we apply incoherent integration to detect the range bin with the highest energy. This bin typically corresponds to the chest movement, as the signal from chest motion tends to produce stronger energy compared to any remaining interference as shown in Fig. 1(c). The expression is as follows:

$$E(j) = \sum_{i=1}^N |X(i, j)| \quad (6)$$

where $E(j)$ denotes the noncoherent integration energy of the j th range bin, $|X(i, j)|$ is the signal amplitude.

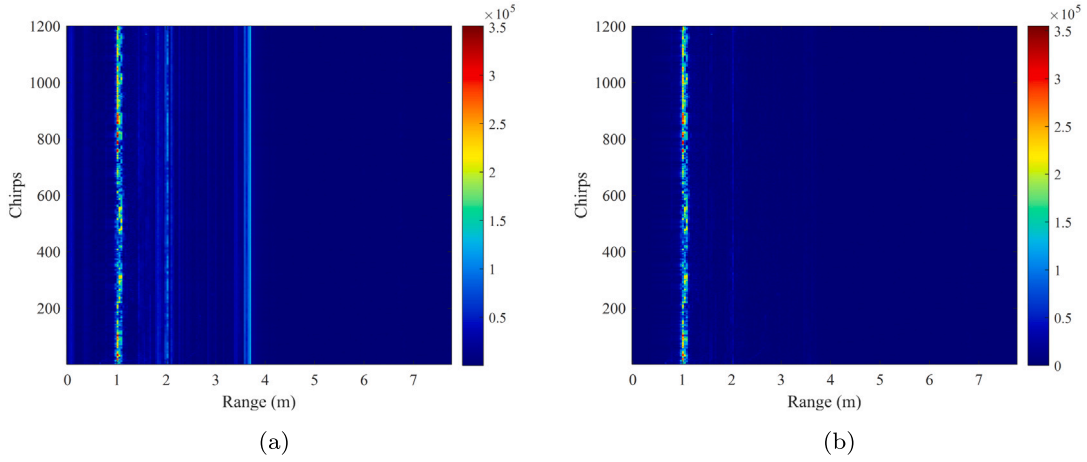


Fig. 3. Comparison of results before and after MTD. (a) Static clutter filter before removal, (b) Static clutter filter after removal.

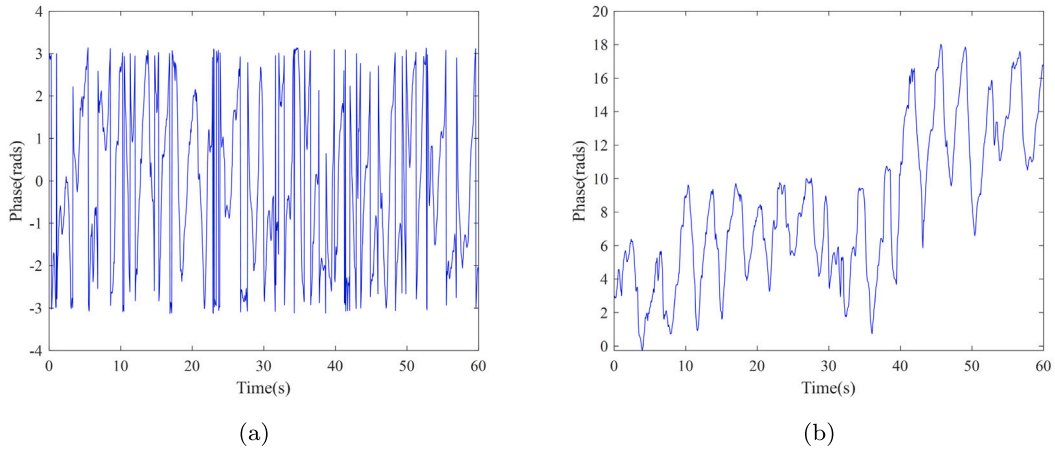


Fig. 4. Examples of arctangent demodulation signal and phase unwrapping signal. (a) Arctangent demodulated signal, (b) Phase unwrapped signal.

Subsequently, phase information is extracted from the target range bin using arctangent demodulation. However, due to the small chest movement amplitude, the phase difference between adjacent mixed signals is typically within the range of $[-\pi, \pi]$ [25]. Exceeding this range causes phase wrapping occurs, as illustrated in Fig. 4(a).

To address this, we perform phase unwrapping. By calculating the phase difference between consecutive frames, points with a phase difference exceeding π from the previous frame are adjusted by $\pm\pi$ to maintain phase continuity. As shown in Fig. 4(b), the unwrapped signal more intuitively reflects human chest movement (i.e., vital signs signals). Therefore, after arctangent demodulation and phase unwrapping, the continuous phase extracted from the baseband signal is expressed as:

$$\phi(t) = \text{unwrap} \left(\arctan \left[\frac{B_Q(t)}{B_I(t)} \right] \right) \quad (7)$$

where the in-phase $B_I(t)$ and the quadrature $B_Q(t)$ components denote the real and imaginary components of the baseband signal, respectively.

3.3. Signal enhancement module

3.3.1. Correlation region selection

In the domain of vital signs signal detection, millimeter-wave technology not only enhances the capability of recognizing spatially proximate targets but also effectively separates overlapping target echoes. Additionally, it significantly suppresses near-field interference signals

via range-domain filtering. However, the human body behaves as an extended target with distributed scattering points. This means that micro-motion information related to vital signs may exist simultaneously across multiple adjacent range bins [26]. Thus, when extracting vital signs signals, it is essential to analyze echo information from multiple range bins comprehensively.

To address this challenge, we employ Pearson correlation coefficients to quantify signal correlation between adjacent range bins. By identifying and processing regions with high phase-frequency consistency to enable targeted signal enhancement. The Pearson correlation coefficient is calculated as follows:

$$\rho(\mu_m, \mu_n) = \frac{\text{cov}(\mu_m, \mu_n)}{\varsigma_{\mu_m} \varsigma_{\mu_n}} \quad (8)$$

where μ_m and μ_n represent the displacement signals currently detected and from the n th adjacent range bin, respectively; $\text{cov}(\mu_m, \mu_n)$ denotes the covariance between these vectors, while ς_{μ_m} and ς_{μ_n} represent the corresponding standard deviations.

Signals with higher correlation coefficients amplify during superposition, while those with lower coefficients attenuate. To leverage this property, we calculate correlations between the target range bin and its four adjacent bins. Signals exceeding a correlation threshold are superimposed to enhance the energy of vital signs signals. Fig. 5(a) illustrates distinct displacement signal variations across range bins corresponding to different chest regions. When the correlation coefficient exceeds a preset threshold (empirically set to 0.8 in this paper [26]), these displacement signals are summed to enhance the vital sign signals as shown in Fig. 5(b). As can be seen from the obtained spectrogram

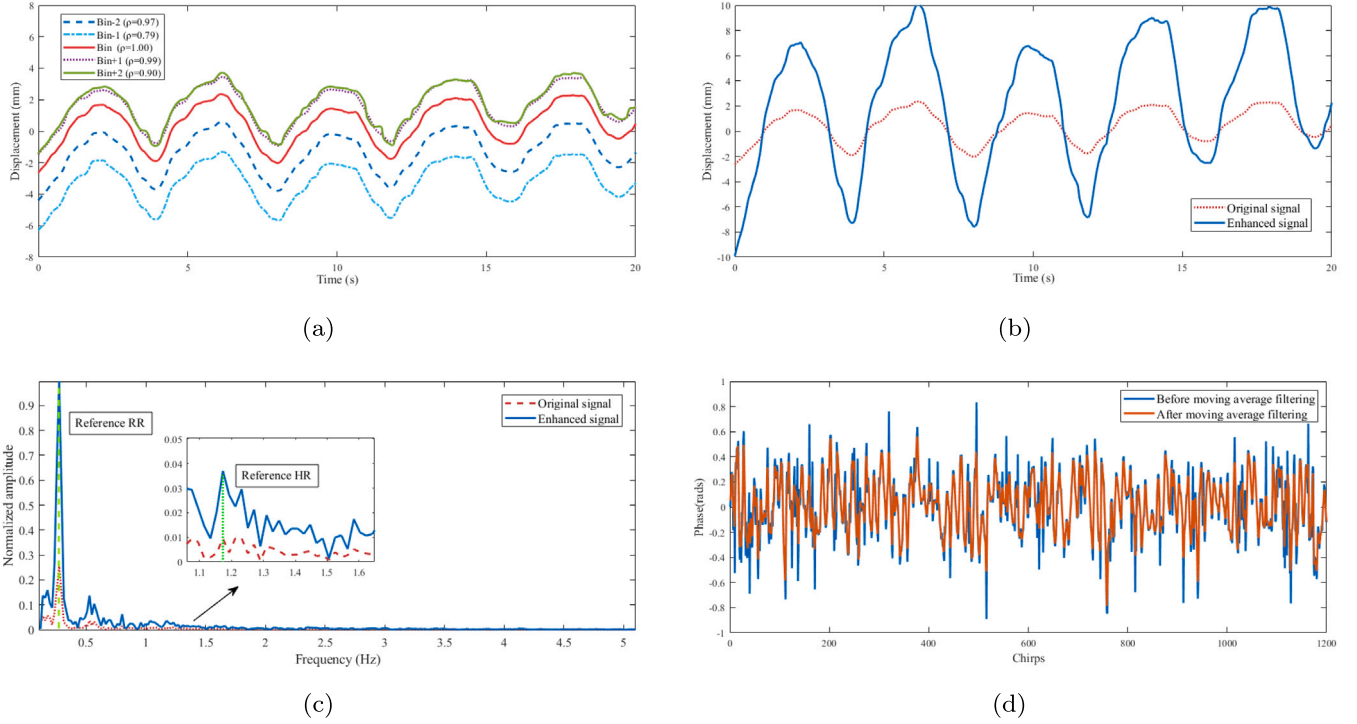


Fig. 5. Result of signal enhancement. (a) Phase demodulation displacement signal of adjacent distance bins, (b) Signal enhancement, (c) Spectrum of original and enhanced signals, (d) Result of Sliding average.

in Fig. 5(c), the heartbeat peaks of the augmented signals are more prominent compared with the original signals.

3.3.2. Phase difference and sliding average filter

Residual phase baseline drift persists even after signal enhancement. To address this, we compute first-order phase differences between consecutive time samples, as follows:

$$\varphi' = [\varphi_2 - \varphi_1, \varphi_3 - \varphi_2, \dots, \varphi_F - \varphi_{F-1}]^T \quad (9)$$

where F denotes the length of the sliding window. This effectively captures phase variations and improves the continuity of heartbeat signals. However, the inherent nature of differencing amplifies high-frequency noise components, causing untreated phase-differenced signals to exhibit significant high-frequency spikes in the time domain [27]. In this paper, we address this issue using a sliding average filter, which suppresses noise by calculating the arithmetic mean of consecutive data points in the signal sequence, as follow:

$$\bar{\lambda}(n) = \frac{1}{N} \sum_{k=0}^{N-1} \varphi(n-k) \quad (10)$$

As shown in Fig. 5(d), comparing the time-domain waveforms before and after filtering, it can be observed that the high-frequency noise components are significantly suppressed.

3.4. Vital signs signal decomposition and reconstruction module

3.4.1. WPD-AC based vital sign separation

In vital signs detection, respiratory signals often significantly interfere with heartbeat signals due to their large amplitude and distinct periodicity. This interference mainly occurs when the harmonics of respiratory signals overlap with heartbeat signals in the frequency domain, making it difficult for traditional filtering techniques to distinguish between them and leading to confusion or errors in heart rate detection. Wavelet packet decomposition (WPD) can divide the signal into components at different scales, each corresponding to a specific frequency range [21]. This multi-resolution characteristic allows

wavelet transforms to capture both the overall trend (low-frequency components) and local details (high-frequency components) of a signal. Given that human respiratory frequencies typically range from [0.2, 0.5] Hz and heart rates from [0.8, 2.0] Hz [28], so we use WPD to extract signal components from different frequency bands, as follows:

$$x(t) = \sum_j c_j \varphi_j(t) + \sum_j \sum_k d_{j,k} \psi_{j,k}(t) \quad (11)$$

Where c_j represents the approximation coefficient, d_j corresponding to the low frequency part; represents the detail coefficient, corresponding to the high frequency part; $\varphi_j(t)$ and $\psi_{j,k}(t)$ are approximate function and wavelet basis function, respectively.

We select the sym4 wavelet basis because it effectively reduces phase distortion during signal decomposition and reconstruction, ensuring the preservation of time-domain characteristics. Based on the target frequency ranges for heart and respiratory signals, the original signal is reconstructed into respiratory and heartbeat signals through wavelet packet decomposition.

Subsequently, to further suppress environmental noise interference in vital signs signals, we employ the autocorrelation function to quantify signal similarity at different time delays. For periodic signals, the autocorrelation function exhibits pronounced peaks at lags corresponding to the signal period, whereas the autocorrelation of noise rapidly decays as the lag increases. This characteristic enables effective enhancement of periodic respiratory and heartbeat signals, as follows:

$$R_{xx}(\tau) = \sum_{n=0}^{N-1-\tau} x(n) \cdot x(n+\tau), \quad \tau \geq 0 \quad (12)$$

where τ denotes the time delay, and $R_{xx}(0) = \sum_{n=0}^{N-1} x(n)^2$ denotes the total energy of the signal when $\tau = 0$. The autocorrelation function is normalized to eliminate the effect of signal amplitude differences, as follows:

$$R_{norm}(\tau) = \frac{R_{xx}(\tau)}{R_{xx}(0)}, \quad \tau \geq 0 \quad (13)$$

After normalization, the range of values of $R_{norm}(\tau)$ is $[-1, 1]$, where $R_{norm}(0) = 1$. The autocorrelation value of random noise decays rapidly

to near zero at $\tau > 0$ with the expression:

$$\lim_{\tau \rightarrow \infty} R_{norm}(\tau) \approx 0 \quad (14)$$

With the introduction of the autocorrelation function, the environmental noise is further suppressed and the peaks associated with respiration rate (RR) and heart rate (HR) dominate.

3.4.2. SBLMS based adaptive signal enhancement

Adaptive filters show significant advantages in processing non-stationary vital signs signals by virtue of real-time adjustment of parameters to adapt to dynamic changes in the input signal and excellent noise rejection. To further improve the detection of respiratory and heartbeat signals, an adaptive signal enhancing method based on the SBLMS algorithm is proposed in this paper.

Given the quasi-periodic nature of respiratory and heartbeat signals, delayed signals retain the inherent waveform and spectral characteristics of the original signal while introducing phase differences through time delays. This allows adaptive filters to more accurately distinguish between valid signals and transient noise during iterative processing, while further reducing the correlation between signal and noise components [29]. Thus, the delayed signal is constructed as the desired signal $d(n)$, expressed as:

$$d(n) = x(n + \varpi_T) \quad (15)$$

where $\varpi_T = 1/f_s$ is the sampling interval, and f_s is the sampling rate.

The output signal $y(n)$ obtained by iterating can be expressed as follows:

$$y(n) = \mathbf{w}^T(n)\mathbf{x}(n) \quad (16)$$

where $\mathbf{w}(n)$ and $\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-Z+1)]^T$ denote the adaptive filter coefficient vector and the input signal vector, respectively, and Z is the filter length. The error expression as follows:

$$e(n) = d(n) - \mathbf{w}^T(n)\mathbf{x}(n) \quad (17)$$

The formula for updating the weight in the traditional LMS algorithm [18] is as follows:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mu \mathbf{x}(n)e(n) \quad (18)$$

where $0 < \mu < 2$ denotes the global step-size, $\mathbf{G}(n)$ is the diagonal matrix of gains.

While effective, conventional implementations exhibit two key limitations: (1) Step-size dependence causing rapid initial convergence followed by stagnation. (2) Performance degradation in sparse signal environments. To address these constraints, we develop a Bayesian sparse Bayesian learning framework. By inferencing the problem through an equivalent observation model with diagonal weighting matrices. The weight update formula for the SBLMS algorithm is derived as [30]:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \ell(n) \frac{e(n)\mathbf{G}(n)\mathbf{x}(n)}{\mathbf{x}^T(n)\mathbf{G}(n)\mathbf{x}(n)} \quad (19)$$

where the expression for variable step-size $\ell(n)$ is given by:

$$\ell(n) = \frac{[\sigma^2(n-1)] \mathbf{x}^T(n)\mathbf{G}(n)\mathbf{x}(n)}{[\sigma^2(n-1)] \mathbf{x}^T(n)\mathbf{G}(n)\mathbf{x}(n) + \delta} \quad (20)$$

where δ denotes the regularization factor, with $\sigma^2(n)$ being given by:

$$\sigma^2(n) = \left[1 - \frac{\alpha(n)\|\mathbf{s}(n)\|^2}{Z} \right] [\sigma^2(n-1)] \quad (21)$$

where $\mathbf{s}(n) = \mathbf{G}^{1/2}(n)\mathbf{x}(n)$, with $\alpha(n)$ is given by:

$$\alpha(n) = \frac{\sigma^2(n-1)}{[\sigma^2(n-1)] \|\mathbf{s}(n)\|^2 + \delta} \quad (22)$$

In order to suppress the sudden pulse interference, the sparse penalty function is introduced, and by adding a weighted paradigm

penalty term to Eq. (19), the weight update formula of the SBLMS algorithm is finally derived as:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \ell(n) \frac{e(n)\mathbf{G}(n)\mathbf{x}(n)}{\mathbf{x}^T(n)\mathbf{G}(n)\mathbf{x}(n)} - \kappa \text{sgn}(\mathbf{w}(n)) \quad (23)$$

where κ denotes the zero attraction coefficient; $\text{sgn}(\cdot)$ denotes the sign function.

Discussion: Considering that vital sign signals in the frequency domain are typically composed of a few dominant components (i.e., respiration and heartbeat) and thus exhibit inherent sparsity, the introduction of sparse Bayesian constraints can effectively suppress noise and non-target components such as respiratory harmonics, thereby enhancing the anti-interference capability of signal reconstruction. Meanwhile, incorporating a zero-attraction term into the weight update process facilitates the rapid shrinkage of small-magnitude irrelevant coefficients toward zero, leading to a clearer spectral structure and more pronounced peaks corresponding to RR and HR, which further improves detection accuracy and stability.

Fig. 6 compares spectrum results before and after SBLMS processing. In Fig. 6(a), both spectra accurately capture the respiratory frequency. However, the unfiltered spectrum contains noticeable interference signals, while the filtered spectrum is more sparse and clear. The contrast in Fig. 6(b) further reveals limitations of the unfiltered approach, the frequency corresponding to the highest amplitude is not the actual heart rate. This is because this frequency band is susceptible to interference from respiratory harmonics and noise, which may cause the system to mistakenly identify respiratory harmonics or noise peaks as the heart rate. The SBLMS-processed spectrum shows significant suppression of respiratory harmonics and noise, making the peaks related to the heart rate more prominent that enable reliable heart rate estimation.

4. System detection experiments

4.1. Experimental setting

In the experiments, we employed the AWR1642BOOST millimeter-wave radar sensor and the DCA1000 EVM data acquisition board, both produced by Texas Instruments (TI), as the hardware modules, as shown in Fig. 7(a). The radar parameters are shown in Table 1. The experiment collected raw data using transmitter Tx1 and receiver Rx2 in a standard indoor office environment, and the transmission power was set to 12 dBm. The data were transmitted to the PC via TI's mmWave Studio software and processed offline using MATLAB. The experimental scenario is shown in Fig. 7(b). To obtain reliable ground-truth measurements, this study employs the HKH-11C respiratory sensor and the HKG-07C infrared pulse sensor, both developed by the Hefei Huake Institute of Electronic Technology, to measure RR and HR, respectively, with sampling rates of 50 Hz and 200 Hz. The system supports high-precision sampling with 8–10 bit resolution and is capable of outputting stable digital physiological waveforms with a high signal-to-noise ratio, thereby ensuring the accuracy and reliability of the experimental reference. Meanwhile, the proposed method employs a six-level wavelet packet decomposition, while the remaining key parameters, including the Pearson correlation threshold, filter length, regularization factor, and zero-attraction coefficient, are summarized in Table 2.

4.2. Experimental results and analysis

To validate the effectiveness of our proposed method, we conducted a comparative analysis with existing techniques from the literature, including bandpass filtering (BPF) [31], complete ensemble empirical mode decomposition (CEEMD) [32], variational mode decomposition (VMD) [21], and improved adaptive parameter variational mode decomposition (IAPVMD) [13]. As demonstrated in Fig. 8(a), while all

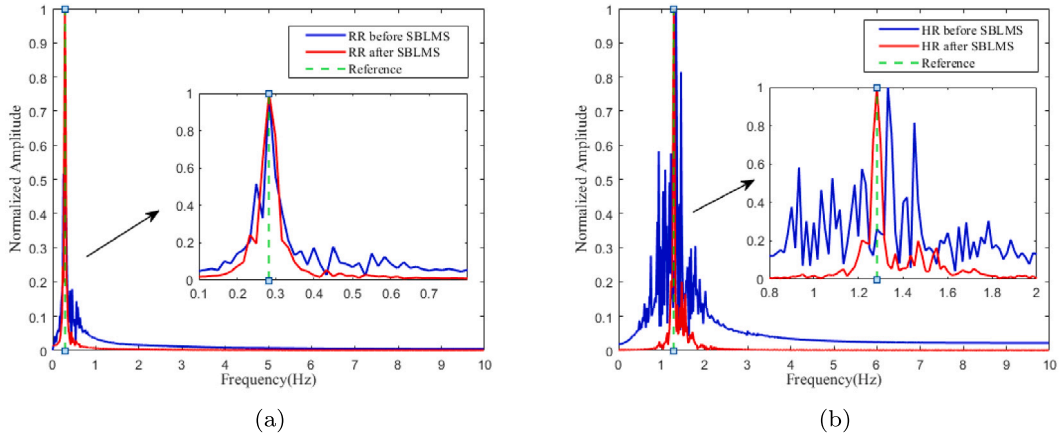


Fig. 6. Results before and after SBLMS processing. (a) RR spectrum and (b) HR spectrum.

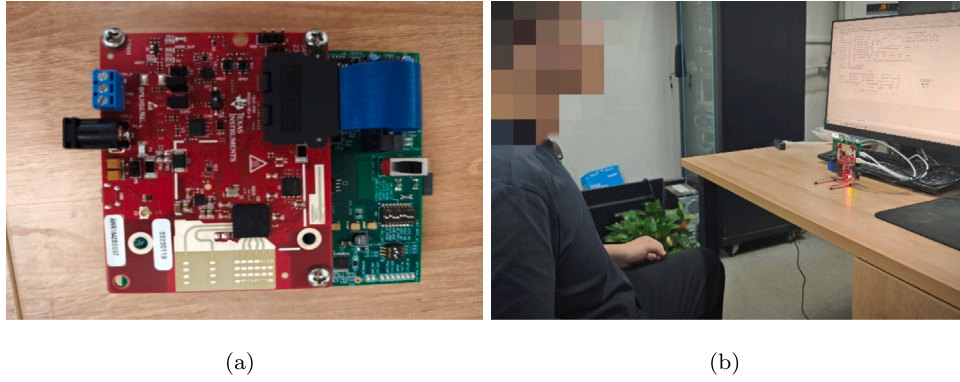


Fig. 7. Experimental Setting. (a) Experimental Hardware and (b) Experimental Scenario.

Table 1
Radar experimental parameter.

Parameter	Value	Parameter	Value
Start Frequency	77 GHz	ADC Samples	200
Frequency Slope	70 MHz/us	ADC Start Time	6 us
Effective Bandwidth	3.99 GHz	Sample Rate	4000 ksps
Idle Time	7 us	Ramp End Time	57 us
Tx Start Time	1 us	Frame Period	50 ms

Table 2
Parameter configuration table.

Parameter	Value	Parameter	Value
ρ	0.8	Z	200
δ	10^{-2}	κ	2×10^{-8}

methods achieved consistent RR estimations, our approach showed superior performance in suppressing spectral clutter and interference compared to conventional techniques. Further observation of Fig. 8(b) reveals that HR detection is susceptible to interference from respiratory harmonics and high-frequency noise, resulting in an inaccurate estimation of heart rate. Although VMD and IAPVMD successfully captured RR values, their outputs exhibited similar susceptibility to the interference. In contrast, the proposed method can effectively suppress the interference of breathing harmonics and high-frequency noise during HR detection, which can significantly improve the detection accuracy.

In addition, to consider the validity of the proposed method, we use the mean absolute error (MAE) and mean absolute percentage error

(MAPE) as measurement metrics, defined as:

$$MAE = \frac{1}{w} \sum_{k=1}^w (H_e(k) - H_t(k)) \quad (24)$$

$$MAEP = \frac{1}{w} \sum_{k=1}^w \frac{(H_e(k) - H_t(k))}{H_t(k)} \quad (25)$$

where w is the total number of time windows during the observation time, $H_e(k)$ and $H_t(k)$ are the estimated value obtained by the proposed algorithm and the true value measured by the sensor in the k th time window, respectively.

The instantaneous RR and HR measurements extracted using our proposed method are shown in Fig. 9. The consistency and reliability of the measurements is shown by the experimental results. In conclusion, the proposed method in this paper shows higher accuracy and robustness in both RR and HR detection, and can effectively reduce the influence of interfering factors.

4.3. Impact of user heterogeneity on RR and HR detection

To evaluate the reliability and accuracy of our proposed method across diverse user populations, we conducted vital signs detection with 7 randomly selected volunteers (3 female and 4 male). During the experiment, participants maintained a stationary seated position 0.8 m directly in front of the FMCW millimeter-wave radar. The detection results are shown in Fig. 10(a) and 10(b) and it can be seen that the average accuracy of RR is 98.28%, and the average accuracy of HR is 99.04%, which are better than other algorithms.

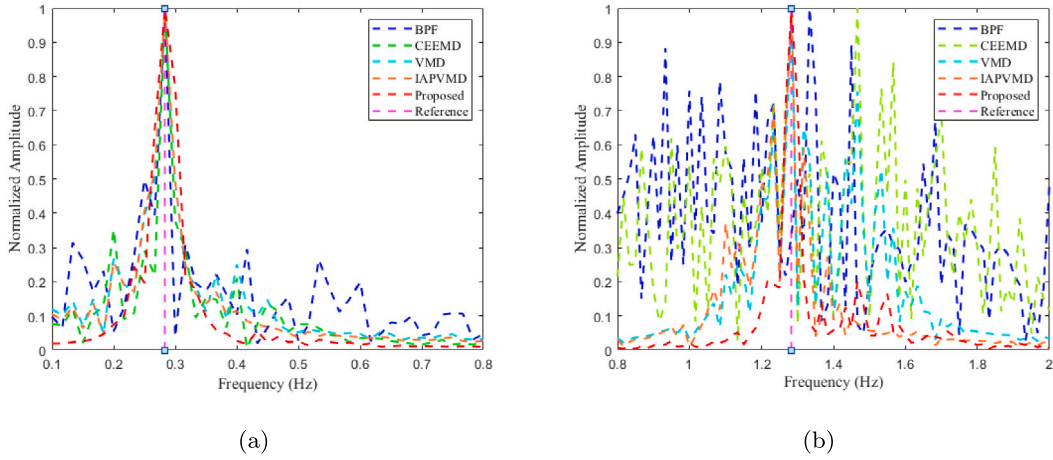


Fig. 8. Experimental results. (a) Estimation result of HR, (b) Estimation result of RR.

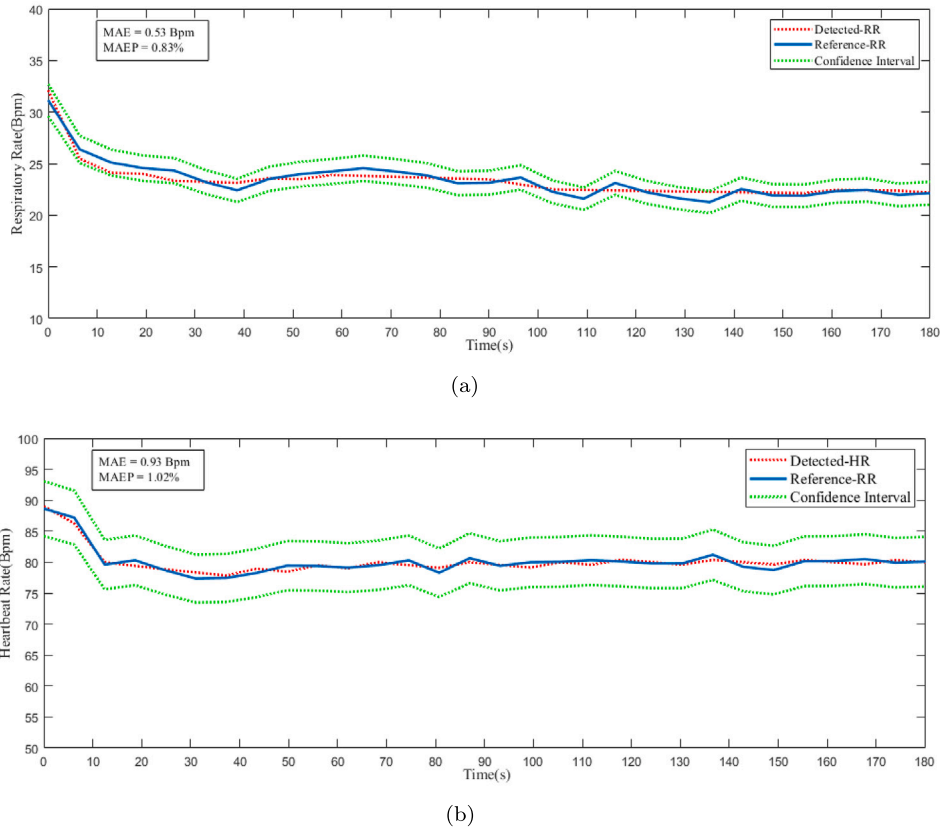


Fig. 9. Results of comparison of real and estimated values. (a) Instantaneous Rate of RR, (b) Instantaneous Rate of HR.

4.4. Effect of different distances on RR and HR detection

In this section, we explore the effect of the distance between the FMCW radar and the human target on the accuracy of RR and HR estimation. The performance of the proposed method at different distances is verified and analyzed in comparison with other methods. For the experimental setup, volunteers faced the radar at different distances from 0.5 m to 2.0 m, as shown in Fig. 11(a).

To further evaluate the detection performance, the statistical results are presented using box plots. In each box plot, the central red dot represents the mean estimation error, while the upper and lower edges of the box represent the third and first quartiles, respectively. The whiskers extend to the maximum and minimum values, excluding outliers displayed separately as individual points. A shorter and narrower

box indicates lower variance in the estimation error and therefore reflects higher algorithm stability. As shown in Fig. 11(b) and 11(c), results demonstrate superior performance at close range, with the proposed method achieving the absolute error of 0.49 BPM for HR estimation and 0.16 BPM for RR measurement at 0.5 m. While signal attenuation and environmental interference in echo signals reduced precision at extended ranges (2.0 m), our method maintained good performance with RR absolute error of 0.7 BPM and HR absolute error of 1.5 BPM. In contrast, conventional methods exhibit more pronounced fluctuations and larger estimation errors as the distance increases. This is primarily due to their limited capability to suppress respiratory harmonics and noise interference, which hampers stable heartbeat extraction in complex environments. By comparison, the proposed system employs an adaptive signal processing architecture

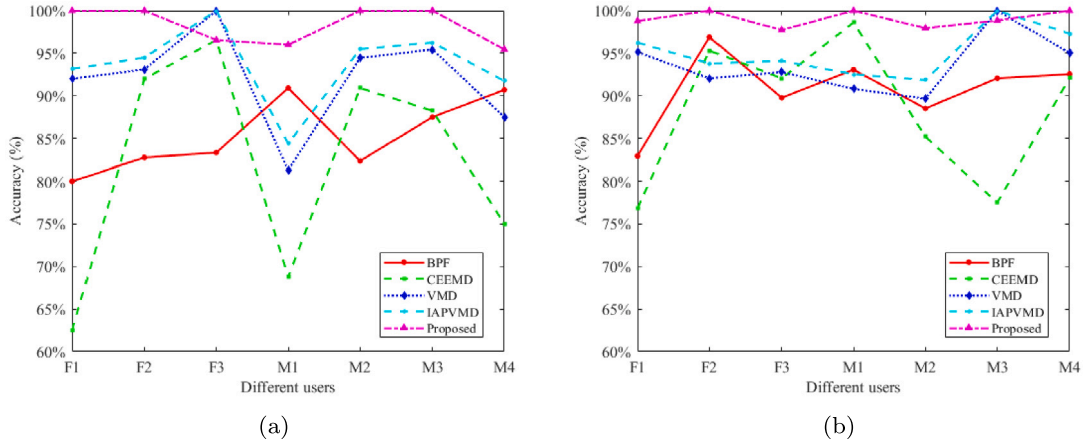


Fig. 10. Experiments on the effect of user heterogeneity. (a) RR accuracy for different users, (b) HR accuracy for different users.

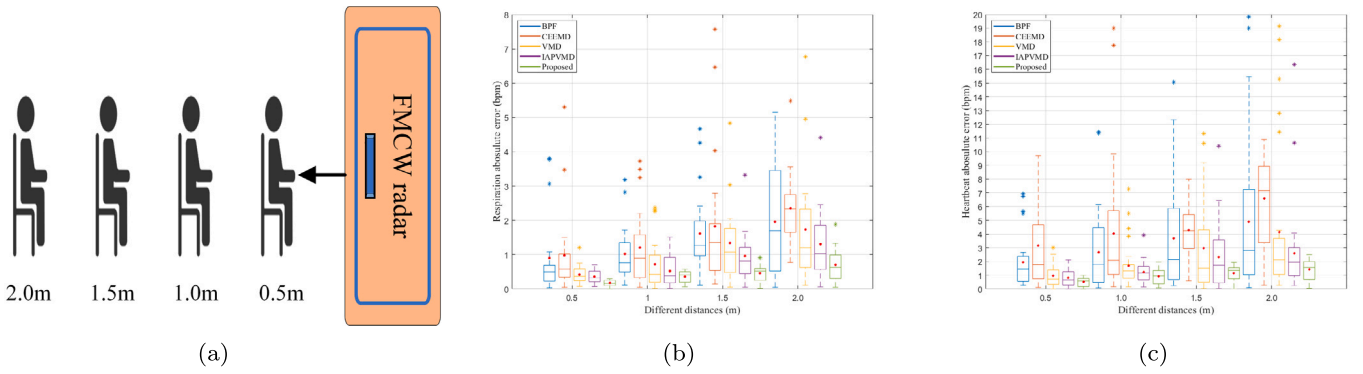


Fig. 11. Experiments on the effect of different distances. (a) Experimental Scenarios, (b) Absolute error of RR estimation, (c) Absolute error of HR.

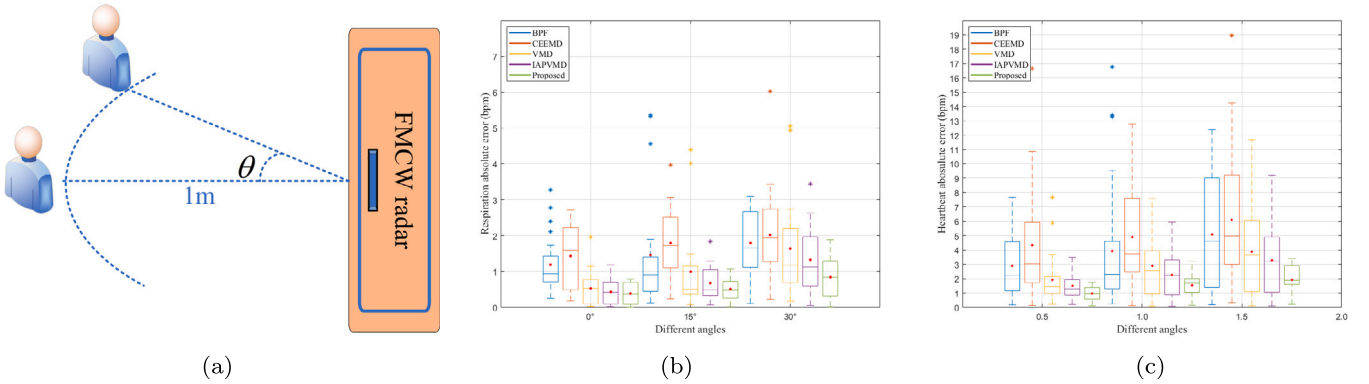


Fig. 12. Experiments on the effect of different angles. (a) Experimental Scenarios, (b) Absolute error of RR estimation, (c) Absolute error of HR.

that effectively enhances interference suppression, thereby enabling reliable vital sign detection.

4.5. Effect of different angles on RR and HR detection

This section examines how angular between volunteers and the FMCW millimeter-wave radar impacts RR and HR detection accuracy. Volunteers were systematically positioned at 1-meter distance facing the radar under three angular configurations (0° , 15° , 30°), as shown in Fig. 12(a).

Similarly, the error distributions at different angles were evaluated using the same box-plot representation. As shown in Fig. 12(b) and 12(c), experimental results demonstrate a progressive increase in estimation error for both RR and HR with greater angular. At 0° , 15° , and

30° , the absolute error of RR measurements gradually increased to 0.38 Bpm, 0.5 Bpm, and 0.84 Bpm, respectively, while the HR estimation errors were similarly 0.95 Bpm, 1.53 Bpm, and 1.92 Bpm, respectively. This degradation correlates with reduced effective reflective surface area and diminished signal-to-noise ratio in radar returns at larger angles, which affects the accuracy of detection. Nevertheless, compared with other conventional methods, the proposed approach demonstrates clear performance advantages across all tested angles. The BPF and CEEMD algorithms exhibit significantly increased estimation errors at 15° and 30° , accompanied by severe extreme errors, indicating that they are highly susceptible to noise and respiratory harmonics under low SNR conditions. Although VMD and IAPVMD show slightly better stability than BPF and CEEMD, their estimation errors still fluctuate noticeably with increasing angles. In contrast, the proposed method

maintains low average errors and high estimation stability across all angles, highlighting its robustness in complex, multi-angle real-world scenarios.

5. Conclusion

In this paper, we present a non-contact vital sign detection system based on FMCW radar that uses a multi-stage signal processing chain to overcome challenges such as extracting weak signals, mitigating respiratory harmonic interference, and suppressing noise in complex settings. In the preprocessing phase, the system applies MTD method to effectively remove static clutter. Next, the vital sign extraction module identifies body areas with concentrated signal energy for preliminary detection. During signal enhancement, combination of Pearson correlation analysis and phase difference techniques strengthens the vital sign signals while reducing baseline drift. Finally, the signal decomposition and reconstruction stage further improve the detection accuracy of respiratory and heartbeat signals using WPD-AC along with SBLMS. Experimental results demonstrate that the system performs robustly in real office environments, consistently maintaining stable performance across varying user conditions, detection distances, and angular configurations which confirming its practical viability. However, detection remains challenging in scenarios with significant body movement (such as walking or limb rotation), which will be the focus of our future research.

Despite the promising performance of the proposed system, several limitations should be acknowledged, particularly in motion scenarios. As revealed by the effect of different angles experiment in Section 4.5, the gradual reduction in the effective radar cross-section of the human thorax leads to weakened echo signals, thereby increasing the difficulty of vital sign detection. This issue is further exacerbated during motion. Movement not only causes persistent instability in the effective radar cross-section but also introduces severe aperiodic phase modulation, resulting in vital sign signals being obscured by complex motion artifacts. To address this limitation, our future research will focus on the iteration and application of motion compensation techniques. By combining the spatial tracking capability of multi-antenna beamforming with the feature restoration capability of deep learning, we aim to effectively suppress motion artifacts and reconstruct echo signals. Alongside optimizing algorithmic performance, we will further explore its real-time deployment on low-power embedded platforms, with the goal of advancing the system toward clinical practicality.

CRedit authorship contribution statement

Pengwei Wen: Writing – review & editing, Writing – original draft, Validation, Software. **Jianpeng Zheng:** Writing – review & editing, Writing – original draft, Software. **Boyang Qu:** Validation, Software. **Xuzhao Chai:** Investigation, Formal analysis. **Li Yan:** Validation, Supervision. **Sheng Zhang:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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